

STUDENT NUMBER _____

NAME: . _____



Trial HSC Mathematics

Extension 1 - 2013

Time Allowed - 2 hours + 5minutes reading

Instructions: Calculators may be used in any parts of the task. For 1 Mark Questions, the correct answer is sufficient to receive full marks. For Questions worth more than 1 Mark, necessary working MUST be shown to receive full marks.

Multiple Choice	/10
Question 11	/15
Question 12	/15
Question 13	/15
Question 14	/15
Total	/70

Fill in the correct answer on the answer sheet - Questions 1 - 10 are worth 1 mark each

1. A parabola has the parametric equations $x = 12t$ and $y = -6t^2$.

The coordinates of the focus are

- | | | | |
|---|-----------|---|-----------|
| A | $(-6, 0)$ | B | $(0, -6)$ |
| C | $(6, 0)$ | D | $(0, 6)$ |

2. If $t = \tan \frac{1}{2}\theta$ then $\frac{1+t^2}{1-t^2}$ is equivalent to

- | | | | |
|---|--------------|---|------------------------------|
| A | $\cot\theta$ | B | $\cos\theta$ |
| C | $\sec\theta$ | D | $\operatorname{cosec}\theta$ |

3. $\frac{\sin(360^\circ - A)}{\sin(90^\circ - A)} =$

- | | | | |
|---|------|---|-----------|
| A | -1 | B | $\tan A$ |
| C | 1 | D | $-\tan A$ |

4. What is the equation of the chord of contact of tangents from $(-2, -1)$ to $x^2 = 4y$

- | | | | |
|---|------------------|---|------------------|
| A | $x + y - 1 = 0$ | B | $x + y + 4 = 0$ |
| C | $x + 2y - 1 = 0$ | D | $x + 2y + 4 = 0$ |

5. Which expression is equal to $\int \sin^2 3x \, dx$?

- | | | | |
|---|--|---|--|
| A | $\frac{1}{2}\left(x - \frac{1}{3}\sin 3x\right) + C$ | B | $\frac{1}{2}\left(x + \frac{1}{3}\sin 3x\right) + C$ |
| C | $\frac{1}{2}\left(x - \frac{1}{6}\sin 6x\right) + C$ | D | $\frac{1}{2}\left(x + \frac{1}{6}\sin 6x\right) + C$ |

6. How many arrangements of the word GEOMETRY are possible if the T and R are to be together in any order?
- A $2 \times 6!$ B $2 \times 7!$
C $7!$ D $\frac{1}{2} \times 7!$
7. Which of the following is an expression for $\int \cos^2 x \sin x \, dx$?
- A $2 \cos x \sin x + c$ B $\cos^3 x + c$
C $\frac{1}{3} \cos^3 x + c$ D $-\frac{1}{3} \cos^3 x + c$
8. What is the acute angle between the lines $y = x + 2$ and $y = 2x - 1$?
- A $18^\circ 26'$ B $19^\circ 28'$
C $70^\circ 32'$ D $71^\circ 34'$
9. If $f(x) = e^{x+2}$ what is the inverse function $f^{-1}(x)$?
- A $f^{-1}(x) = e^{y-2}$ B $f^{-1}(x) = e^{y+2}$
C $f^{-1}(x) = \log_e x - 2$ D $f^{-1}(x) = \log_e x + 2$
10. What is the derivative of $y = \cos^{-1}\left(\frac{1}{x}\right)$ with respect to x ?
- A $\frac{-1}{\sqrt{x^2-1}}$ B $\frac{-1}{x\sqrt{x^2-1}}$
C $\frac{1}{\sqrt{x^2-1}}$ D $\frac{1}{x\sqrt{x^2-1}}$

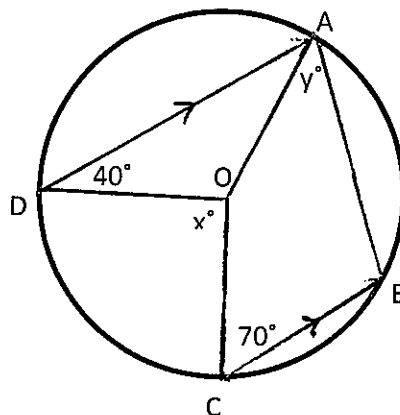
- a) The point P divides the interval AB joining A $(-4, -3)$ and B $(1, 5)$ externally in the ratio 3:2. Find the coordinates of P. 2
- b) At a dinner party, the host, hostess and their six guests sit at a round table. In how many ways can they be arranged if the host and hostess are separated? 2
- c) A stone is thrown from the top of a 15m cliff with an initial velocity of $26ms^{-1}$ at an angle of projection equal to $\tan^{-1}\left(\frac{5}{12}\right)$ above the horizontal.
- (i) Taking $g = -10$, derive equations for x and y in terms of t . 3
- (ii) Calculate the time when the stone will reach the ground. 2
- d) Solve $\frac{x^2-4}{2x} < 0$ and graph your solution on a number line. 3
- e) Differentiate $y = x \sin^{-1} x + \sqrt{1-x^2}$ 3
- Hence evaluate $\int_0^1 \sin^{-1} x \, dx$.

Question 12 15 Marks **(Begin a new sheet of paper)**

Marks

- a) O is the centre of the circle and AD is parallel to BC.
Find the values of x and y .

3



- b) The acceleration of a particle is defined in terms of its position by the equation $a = 2x + 4$. If $v = 5$ when $x = 2$, what is the velocity when $x = 4$?

3

- c) The normal to a parabola $x^2 = 4ay$ at a point $P(2ap, ap^2)$ on it cuts the axis of the parabola at G.

(i) Show that the normal has equation $x + py = 2ap + ap^3$.

(ii) Find the coordinates of G

2

(iii) Find the equation of the locus of the mid-point M of PG

1

2

- d) Cane sugar, when placed in water, converts to dextrose at a rate proportional to the amount of unconverted material remaining. That is, if M grams is the amount of material converted after t minutes, then

$$\frac{dM}{dt} = k(S - M) \quad \text{where } S \text{ grams is the initial amount of cane sugar and } k \text{ is a constant.}$$

(i) Show that $M = S + Ae^{-kt}$ satisfies the equation, where A is constant.

(ii) If a certain amount of cane sugar is placed in water at time $t = 0$ and 40% of it has been converted after 10 minutes, show $k = \frac{1}{10} \log_e \frac{5}{3}$.

1

2

- a) A particle is oscillating in simple harmonic motion about a fixed point O on a straight line. At time t seconds its displacement x metres from O satisfies the equation

$$\frac{d^2x}{dt^2} = -4x$$

- (i) Show that $x = a \cos(2t + \beta)$ is a possible equation of motion for this particle, where a and β are constants and $a > 0$. State the period. 2
- (ii) The particle is observed at time $t = 0$ to have a velocity of 2 metres per second and a displacement from the origin of 4 metres. Show that the amplitude of the oscillation is $\sqrt{17}$ metres. 2
- (iii) Show that β is approximately -0.245 correct to 3 decimal places. 1
- (iv) Give the general solutions to the times when the particle is back at its starting point. 2
- (v) Find the time at which the particle first returns to its starting point. 1
- b) By using the substitution $u = \sqrt{x+1}$ find $\int_3^8 \frac{x-1}{\sqrt{x+1}} dx$ 3
- c) Use the remainder theorem to find one factor of $x(x+1) - a(a+1)$. By division, or otherwise, find the other factor. 2
- d) It is known that the equation $e^{-x^2} - 5x^2 - 0.99 = 0$ has a positive root close to the origin. Attempt to find this root using Newton's method, starting with a first approximation of $x_0 = 0$. Explain why this method fails. 2

a) Prove by induction that

$$\frac{1}{x^n(x-1)} = \frac{1}{x-1} - \frac{1}{x} - \frac{1}{x^2} - \dots - \frac{1}{x^n} \quad 4$$

for all positive integers n where $x \neq 0$ or 1 .

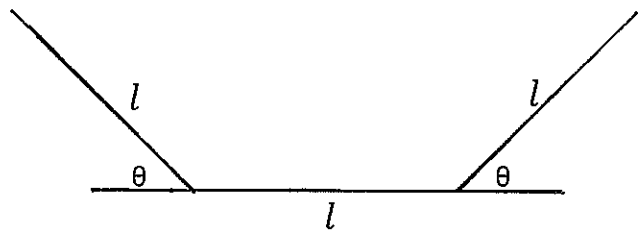
(Hint: When $n = 1$, RHS contains two terms.)

b) Let α, β and γ be the roots of $3x^3 + 8x^2 - 1 = 0$.

(i) State the values of $\alpha + \beta + \gamma$, $\alpha\beta + \beta\gamma + \gamma\alpha$, and $\alpha\beta\gamma$. 1

(ii) Find the value of $\left(\alpha + \frac{1}{\beta}\right)\left(\beta + \frac{1}{\gamma}\right)\left(\gamma + \frac{1}{\alpha}\right)$ 3

c) An irrigation channel is to have a cross-section in the shape of a trapezium. The bottom and sides are each l metres long. The sides of the channel make an angle $\theta \leq \frac{\pi}{2}$ with the horizontal.



(i) Show that the area of cross-section of the channel, as a function of θ , is 3
 $A = l^2 \sin \theta (1 + \cos \theta)$ (l is a constant).

(ii) For what angle θ is the area of the cross-section a maximum? 4

End of Test

Solutions to 2013 Ext 1 Trial.

1. B $(0, -6)$ 

2. $\cos \theta = \frac{1-t^2}{1+t^2} \quad \therefore \frac{1+t^2}{1-t^2} = \sec \theta$ C

3. $\frac{\sin(360-A)}{\sin(90-A)} = \frac{-\sin A}{\cos A} = -\tan A$ D

4. $xx_0 = 2a(y+y_0) \quad a=1$

$$-2x = 2(y-1)$$

$$0 = 2x + 2y - 2$$

$$0 = x + y - 1$$

A

5. $\cos 2\theta = 1 - 2\sin^2 \theta$

$$2\sin^2 \theta = 1 - \cos 2\theta \Rightarrow \sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$$

$$\int \sin^2 3x \, dx = \frac{1}{2} \int (1 - \cos 6x) \, dx$$

$$= \frac{1}{2} \left(x - \frac{1}{6} \sin 6x \right)$$

C

6. Bind them together 2! ways

Arrange 7 things 7!

$$2 \text{ are alike} \quad \therefore \frac{2! \times 7!}{2!} = 7! \quad C$$

7. $\int \cos^2 x \sin x \, dx = \int -\cos^2 x (-\sin x) \, dx$

$$= -\frac{\cos^3 x}{3} + C$$

D

8. $m_1 = 1 \quad m_2 = 2 \quad \tan \theta = \left| \frac{1-2}{1+2} \right|$

$$\tan \theta = \frac{1}{3} \quad \theta = 18^\circ 26'$$

A

9. $y = e^{x+2}$

$$x = e^{y+2}$$

$$y+2 = \ln x$$

$$y = \ln x - 2$$

C

$$y = \cos^{-1} \frac{1}{x}$$

$$y' = \frac{-1}{\sqrt{1 - \frac{1}{x^2}}} \cdot -x^{-2}$$

$$y' = \frac{1}{x^2 \sqrt{1 - \frac{1}{x^2}}} = \frac{1}{x \sqrt{x^2 - 1}} \quad (1)$$

(Note: $\frac{1}{x} = \cos y \Rightarrow x = \frac{1}{\cos y} = \sec y$)

$\therefore y = \sec^{-1} x$ We have differentiated $y = \sec^{-1} x$

Q11

a) $x = \frac{kx_2 + lx_1}{k+l}$

$y = \frac{ky_2 + ly_1}{k+l}$

$\frac{k}{l} = \frac{3}{-2}$

$x = \frac{3 \times 1 + -2 \times -4}{3 - 2}$

$y = \frac{3 \times 5 + -2 \times -3}{3 - 2}$

$x = 11$

$y = 21$

$P(11, 21)$

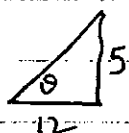
b) 8 people. Put host + hostess together 2 ways
Arrange other 6 $6!$ ways

$\therefore 2 \times 6!$ with host + hostess together.

$\therefore$ Separated $7! - 2 \times 6! = 3600$ ways

c) $V = 26$ $\theta = \tan^{-1} \frac{5}{12}$

Initially



$\dot{y} = \frac{5}{13} \times 26 = 10$

15m

$\dot{x} = \frac{12}{13} \times 26 = 24$

$\therefore \ddot{x} = 0$

$\ddot{y} = -10$

$\dot{x} = 24$

$\dot{y} = -10t + 10$

$x = 24t$

$y = -5t^2 + 10t + 15$

ii) $y = 0$

$-5t^2 + 10t + 15 = 0$

$t^2 - 2t - 3 = 0$

$(t - 3)(t + 1) = 0$

$t = 3$ or -1

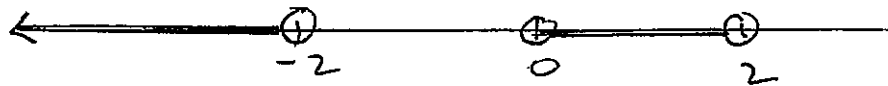
$\therefore$ Stone hits ground after 3 seconds.

11 d) $\frac{x^2 - 4}{2x} < 0$

Consider $x > 0$
 Solve $x^2 - 4 < 0$
 $(x-2)(x+2) < 0$
 $0 < x < 2$

Consider $x < 0$
 Solve $x^2 - 4 > 0$
 $(x-2)(x+2) > 0$
 $x < -2$

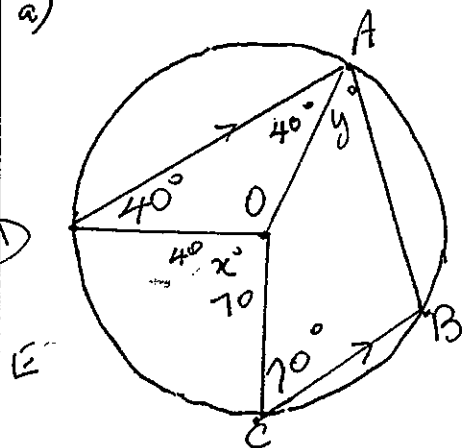
$\therefore 0 < x < 2 \quad \text{or} \quad x < -2$



e) $y = x \sin^{-1} x + \sqrt{1-x^2}$
 $y' = x \cdot \frac{1}{\sqrt{1-x^2}} + \sin^{-1} x - 1 + \frac{1}{2} (1-x^2)^{-1/2} \cdot -2x$
 $= \frac{x}{\sqrt{1-x^2}} + \sin^{-1} x - \frac{x}{\sqrt{1-x^2}} = \sin^{-1} x$
 $\therefore \int_0^1 \sin^{-1} x \, dx = \left[x \sin^{-1} x + \sqrt{1-x^2} \right]_0^1$
 $= \sin^{-1} 1 + 0 - 0 - \sqrt{1}$
 $= \frac{\pi}{2} - 1$

Q 12

a)



$\angle DAO = 40^\circ$ (isosceles $\triangle O D = O A$)

Draw $O E \parallel A D$

$\therefore \angle D O E = 40^\circ$ (alternate $\angle$'s)

$\angle E O C = 70^\circ$ (alternate $\angle$'s in $\parallel$ lines)

$\therefore x = 110^\circ$

$\angle A O D = 100^\circ$ (angle sum of $\triangle$)

$\therefore \angle A B C = \frac{1}{2} (100 + 110)$

$\angle A B C = 105^\circ$ (Angle at centre

is twice $\angle$ at circumference standing on major arc AC.

Also $\angle A O C = 360 - (100 + 110) = 150^\circ$

$\therefore y = 360 - (70 + 150 + 105)$ angle sum of quad.
 $y = 25^\circ$

12b)

$$\ddot{x} = 2x + 4$$

$$\ddot{x} = \frac{d(\frac{1}{2}v^2)}{dx}$$

$$\therefore \frac{1}{2}v^2 = \int (2x+4) dx$$

$$\frac{1}{2}v^2 = x^2 + 4x + C_1$$

$$v^2 = 2x^2 + 8x + C$$

$$v=5, x=2$$

$$\therefore 25 = 8 + 16 + C$$

$$1 = C$$

$$v^2 = 2x^2 + 8x + 1$$

$$\text{when } x=4,$$

$$v^2 = 32 + 32 + 1$$

$$v^2 = 65$$

$$v = \pm \sqrt{65}$$

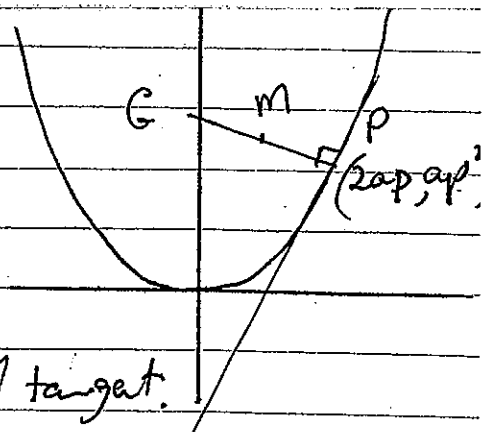
But $v > 0$ when $x=2$ & $a > 0$ when $x=2$

$$\therefore v = \sqrt{65}$$

c) $x = 2ap$
 $y = ap^2$

$$\frac{dy}{dx} = \frac{\frac{dy}{dp}}{\frac{dx}{dp}}$$

$$= \frac{2ap}{2a} = p = \text{Gradient of tangent.}$$



Gradient of normal is $-\frac{1}{p}$

$$\therefore \text{Normal is } y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap$$

$$x + py = 2ap + ap^3$$

provided $p \neq 0$, this cuts $x=0$ at G

$$py = 2ap + ap^3$$

$$y = 2a + ap^2$$

$$\therefore G \text{ is } (0, 2a + ap^2)$$

$$M \text{ is } x = \frac{2ap + 0}{2} = ap$$

$$y = \frac{ap^2 + 2a + ap^2}{2} = a + ap^2$$

$$p = \frac{x}{a} \therefore y = a + a\left(\frac{x^2}{a^2}\right)$$

$$y - a = \frac{x^2}{a}$$

$$\text{or } x^2 = a(y - a) \text{ with } x=0 \text{ excluded.}$$

This is a parabola with vertex $(0, a)$ + focal length $\frac{a}{4}$. However, vertex $(0, a)$ must be excluded from locus.

$$d) \quad \frac{dM}{dt} = k(S - M)$$

$$i) \quad M = S + Ae^{-kt}$$

$$\frac{dM}{dt} = 0 + A e^{-kt} \cdot -k$$

$$= -k A e^{-kt}$$

But

$$M - S = A e^{-kt}$$

$$\therefore \frac{dM}{dt} = -k(M - S)$$

$$= k(S - M) \text{ as required.}$$

$$ii) \quad t=0, M=0$$

$$M = S + Ae^{-kt}$$

$$0 = S + A \Rightarrow A = -S$$

$$t=10, M=40$$

$$S = 100$$

(%)

$$40 = 100 - 100 e^{-10k}$$

$$100 e^{-10k} = 60$$

$$e^{-10k} = \frac{3}{5} \Rightarrow e^{10k} = \frac{5}{3}$$

$$\therefore 10k = \ln \frac{5}{3}$$

$$k = \frac{1}{10} \ln \frac{5}{3}$$

as required

$$\therefore M = S + A e^{-\frac{1}{10} \ln \frac{5}{3} t}$$

13

a) $\frac{d^2x}{dt^2} = -4x$

i) $x = a \cos(2t + \beta)$

$$\frac{dx}{dt} = v = -2a \sin(2t + \beta)$$

$$\frac{d^2x}{dt^2} = \ddot{x} = -4a \cos(2t + \beta)$$

$$= -4x \quad \text{as required}$$

$$\text{Period} = \frac{2\pi}{2} = \pi$$

ii) $t=0, v=2, x=4$

$$4 = a \cos \beta \Rightarrow \cos \beta = \frac{4}{a}$$

$$2 = -2a \sin \beta \Rightarrow \sin \beta = -\frac{1}{a}$$

$$\sin^2 \beta + \cos^2 \beta = 1$$

$$\frac{1}{a^2} + \frac{16}{a^2} = 1$$

$$\frac{17}{a^2} = 1 \Rightarrow a = \sqrt{17} \quad (a > 0)$$

iii) $\tan \beta = \frac{-\frac{1}{a}}{\frac{4}{a}} = -\frac{1}{4}$

$\cos$ +ve $\sin$ -ve $\therefore$ 4th quad for β .

$$\beta = \tan^{-1}\left(-\frac{1}{4}\right) \approx -0.245$$

iv) $x=4 \quad x = \sqrt{17} \cos(2t - 0.245)$

$$4 = \sqrt{17} \cos(2t - 0.245)$$

$$\cos(2t - 0.245) = \frac{4}{\sqrt{17}}$$

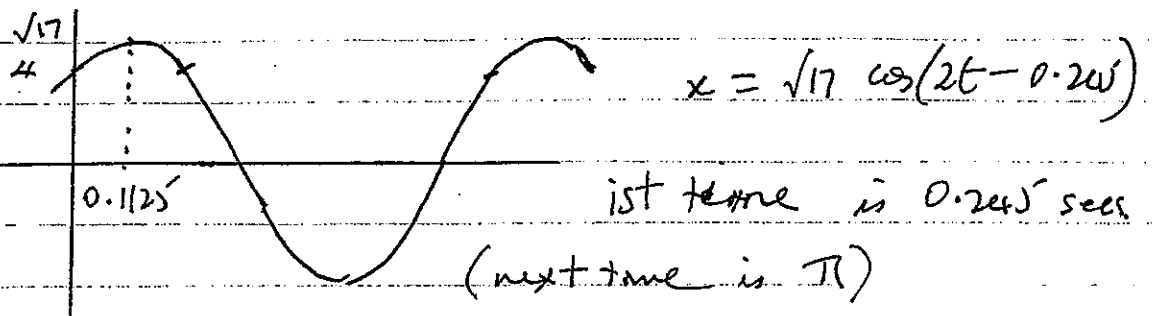
$$2t - 0.245 = 2n\pi \pm \cos^{-1} \frac{4}{\sqrt{17}}$$

$$2t - 0.245 = 2n\pi \pm 0.245$$

$$2t = 2n\pi + 2 \times 0.245 \quad \text{or} \quad 2n\pi$$

$$t = n\pi + 0.245 \quad \text{or} \quad n\pi$$

✓) $n=0, t=-\beta = 0.245$



b) $\int_3^8 \frac{x-1}{\sqrt{x+1}} dx$

$u = \sqrt{x+1}$
 $u^2 = x+1$ $x = u^2 - 1$
 $2u du = dx$

$x=3, u=2$
 $x=8, u=3$

$= \int_2^3 \frac{u^2 - 2}{u} \cdot 2u du$

$= 2 \int_2^3 (u^2 - 2) du$

$= 2 \left[\frac{u^3}{3} - 2u \right]_2^3$

$= 2 \left(\frac{27}{3} - 6 - \frac{8}{3} + 4 \right)$

$= 8\frac{2}{3}$

c) $P(x) = x(x+1) - a(a+1)$
 $P(a) = a(a+1) - a(a+1)$
 $= 0$

$\therefore x-a$ is a factor.

otherwise method:

$x^2 + x - a^2 - a$
 $(x^2 - a^2) + (x - a)$
 $(x-a)(x+a) + (x-a)$

$P(x) = (x-a)(x+a+1)$

$$d) \quad f(x) = e^{-x^2} - 5x^2 - 0.99$$

$$\begin{aligned} f'(x) &= e^{-x^2} \cdot -2x - 10x \\ &= -2x e^{-x^2} - 10x \\ &= -2x \{e^{-x^2} + 5\} \end{aligned}$$

$$f(0) = 1 - 0.99 = 0.01$$

$$f'(0) = 0$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 0 - \frac{0.01}{0}$$

Method fails because there is a stationary point at $x=0$.
 $\therefore$ Tangent at $x=0$ does not cross x axis for next approximation.

Q 14

$$\frac{1}{x-1} - \frac{1}{x} - \frac{1}{x^2} - \dots - \frac{1}{x^n} = \frac{1}{x^n(x-1)}$$

① Prove true for $n=1$

$$LHS = \frac{1}{x-1} - \frac{1}{x} = \frac{x - (x-1)}{x(x-1)} = \frac{1}{x(x-1)}$$

$$RHS = \frac{1}{x^1(x-1)} \quad \text{as required}$$

$\therefore$ true for $n=1$

② Assume true for $n=k$

$$\text{i.e. } \frac{1}{x-1} - \frac{1}{x} - \frac{1}{x^2} - \dots - \frac{1}{x^k} = \frac{1}{x^k(x-1)} \quad \rightarrow *$$

③ Consider $\frac{1}{x-1} - \frac{1}{x} - \frac{1}{x^2} - \dots - \frac{1}{x^k} - \frac{1}{x^{k+1}}$

$$= \frac{1}{x^k(x-1)} - \frac{1}{x^{k+1}} \quad \text{using } *$$

$$= \frac{x - (x-1)}{x^{k+1}(x-1)} = \frac{1}{x^{k+1}(x-1)}$$

which is of the form $\frac{1}{x^n(x-1)}$ when
 $n = k+1$.

(v) Since statement is true for $n=1$
 and if true for $n=k$, is also true
 for $n=k+1$, then it is true
 for all positive integers n .

b) $3x^3 + 8x^2 - 1 = 0$

i) $\Sigma \alpha = \alpha + \beta + \gamma = -\frac{8}{3} \quad \left(-\frac{b}{a}\right)$

$\Sigma \alpha\beta = \alpha\beta + \beta\gamma + \gamma\alpha = 0 \quad \left(\frac{c}{a}\right)$

$\alpha\beta\gamma = -\frac{d}{a} = \frac{1}{3}$

ii) $\left(\alpha + \frac{1}{\beta}\right)\left(\beta + \frac{1}{\gamma}\right)\left(\gamma + \frac{1}{\alpha}\right) =$

$\alpha\beta\gamma + \alpha\beta \cdot \frac{1}{\alpha} + \alpha \frac{1}{\gamma} \cdot \gamma + \alpha \frac{1}{\gamma} \frac{1}{\alpha} + \frac{1}{\beta} \beta\gamma +$

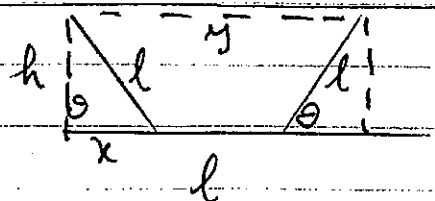
$\frac{1}{\beta} \cdot \beta \cdot \frac{1}{\alpha} + \frac{1}{\beta} \cdot \frac{1}{\gamma} \cdot \gamma + \frac{1}{\beta} \cdot \frac{1}{\gamma} \cdot \frac{1}{\alpha}$

$= \alpha\beta\gamma + (\alpha + \beta + \gamma) + \left(\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}\right) + \frac{1}{\alpha\beta\gamma}$

$= \alpha\beta\gamma + (\alpha + \beta + \gamma) + \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} + \frac{1}{\alpha\beta\gamma}$

$= \frac{1}{3} - \frac{8}{3} + 0 + 3 = \frac{2}{3}$

c) $\frac{h}{l} = \sin \theta$
 $h = l \sin \theta$



$\frac{x}{l} = \cos \theta$
 $x = l \cos \theta$

$\therefore y = 2x + l = 2l \cos \theta + l$

$$Area = \frac{l}{2} (a+b) = \frac{l \sin \theta}{2} (2l \cos \theta + l + l)$$

$$Area = \frac{l \sin \theta}{2} \{ 2l \cos \theta + 2l \}$$

$$A = l^2 \sin \theta (\cos \theta + 1) \quad \text{as required}$$

$$\begin{aligned} \text{ii) } \frac{dA}{d\theta} &= l^2 \{ \sin \theta (-\cos \theta) + (\cos \theta + 1) \sin \theta \} \\ &= l^2 \{ -\sin^2 \theta + \cos^2 \theta + \cos \theta \} \\ &= l^2 \{ \cos^2 \theta - 1 + \cos^2 \theta + \cos \theta \} \\ &= l^2 \{ 2 \cos^2 \theta + \cos \theta - 1 \} \end{aligned}$$

$$\frac{dA}{d\theta} = 0 \quad \text{for maximum or minimum area.}$$

$$2 \cos^2 \theta + \cos \theta - 1 = 0$$

$$(2 \cos \theta - 1)(\cos \theta + 1) = 0$$

$$\cos \theta = \frac{1}{2} \quad \text{or} \quad \cos \theta = -1$$

$$\theta = \frac{\pi}{3}$$

out of range of θ

Show that Area is in fact maximum.

$$\frac{d^2A}{d\theta^2} = l^2 \{ 4 \cos \theta (-\sin \theta) - \sin \theta \}$$

$$= -l^2 \{ 4 \cos \theta \sin \theta + \sin \theta \}$$

when $\theta = \frac{\pi}{3}$, $4 \cos \theta \sin \theta + \sin \theta$ is positive

$\therefore \frac{d^2A}{d\theta^2}$ is negative

$\therefore A$ concave down

$\therefore$ Area is maximum.

$\therefore$ Area is maximum when $\theta = \frac{\pi}{3}$